SAURABH PATHAK

AI Engineer | Full-Stack Developer (MERN, Next.js) | AI Agents and MCP

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PROFESSIONAL SUMMARY

AI Engineer and Full-Stack Developer with 2+ years of production experience building scalable web applications and AI agent infrastructure. Built Bo2bot, an agent-to-agent messaging platform on REST APIs and Model Context Protocol (MCP), and open-sourced mcp-safe-actions, a tested human-approval library for MCP tools. Strong in the MERN stack, Next.js, TypeScript, Python, JWT and RBAC, multi-agent workflows, LLM applications and Retrieval-Augmented Generation (RAG); ships end to end with Docker, CI/CD and SEO.

TECHNICAL SKILLS

AI and LLM: Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Model Context Protocol (MCP), AI Agents, Multi-Agent Workflows, Agent-to-Agent Messaging, Human-in-the-Loop Approval, Prompt Engineering, OpenRouter, OpenClaw, Hermes, Token-Cost Optimization
Languages: JavaScript (ES2023+), TypeScript, Python, C++, PHP, HTML5, CSS3
Frontend: React.js, Next.js, Tailwind CSS, Responsive Design, Figma
Backend and APIs: Node.js, Express.js, REST APIs, JWT and Bearer-Token Authentication, RBAC, Microservices
Databases: MongoDB, MySQL, Firebase
Cloud, DevOps and Testing: Google Cloud Platform (GCP), Microsoft Azure, Docker, Docker Compose, Git, GitHub, GitHub Actions, CI/CD, pytest, Node.js test runner
SEO and Analytics: Technical SEO, Structured Data, Core Web Vitals, Google Analytics

PROFESSIONAL EXPERIENCE

Software Developer (Full-Stack) | Sprilio Tech | Oct 2024 - Present

- Build and ship full-stack web applications using React.js, Next.js, Node.js, Express.js and MongoDB.
- Designed RESTful APIs with JWT authentication and role-based access control (RBAC); cut average API response time by 35% through query optimization and indexing.
- Designed and deployed scalable microservices using Docker Compose, with real-time data sync via Firebase.
- Led technical SEO (structured data, Core Web Vitals, on-page optimization), driving 3x organic traffic growth and a 60% improvement in page load scores.

PROJECTS

mcp-safe-actions - Human Approval for AI Agent Tools | Open Source Author | 2026
github.com/techsaurabh-394/mcp-safe-actions | TypeScript, Node.js, Model Context Protocol (MCP), zod, GitHub Actions

- Built an open-source TypeScript library that makes AI agents wait for human approval before running destructive MCP tools.
- Two approval paths: MCP elicitation prompts the user directly (the model cannot approve itself); a fallback issues a one-time, expiring token tied to the exact arguments. Errors and timeouts never run the action; the audit log never stores raw arguments.
- 49 automated tests (unit, property-based, end-to-end over stdio), 100% line coverage and a mutation check of 10 injected bugs, all caught; CI on Node 18, 20 and 22 with GitHub Actions. Found and fixed a symlink path-traversal flaw.

Bo2bot - Agent-to-Agent Messaging Platform | Full-Stack and AI Developer | 2026 - Present
bo2bot.com | REST API, Model Context Protocol (MCP), bearer-token authentication, AI agents

- Built a messaging platform where each AI agent has its own address and can message agents on other platforms (OpenClaw, Hermes, Claude Desktop) without a human relay.
- Designed a token-efficient REST API (bearer-token REST or MCP, no SDK) with relationship-based inbox sorting, a six-verdict reputation score and a public bulletin board for agent discovery.

Rahidera - Community Travel Marketplace | Full-Stack Developer | Nov 2025 - Present
rahidera.com | full-stack web application, KYC verification, escrow payments, OTP, technical SEO

- Designed, built and launched a full-stack travel marketplace from zero to production where hosts list experiences and travellers search, filter and book trips; onboarded 50+ early hosts and users in month one.
- Built KYC identity verification (document and selfie matching), escrow payments and OTP-confirmed trip handoffs; drove 500+ organic visits through technical SEO and community outreach.

EDUCATION

B.Tech, Computer Science and Engineering | ABES Institute of Technology (AKTU) | 2020 - 2024 | CGPA: 8.0/10

CERTIFICATIONS AND ACHIEVEMENTS

- Meta Full-Stack Developer Certificate (Coursera); AWS Cloud Essentials Certificate (Coursera)
- Runner-Up, Smart India Hackathon 2022; 200+ competitive programming problems solved
- 200+ technical articles on React, Node.js and AI engineering, 10K+ monthly readers (mernstackdev.com)